

In [1]: `import pandas as pd`

In [7]: `# TASK 1 - IPL total runs by team`

```
ipl = pd.read_csv("ipl_team_runs.csv")

print(ipl.groupby("team")["runs"].sum().to_frame("total_runs"))
```

total_runs	
team	
CSK	550
MI	540
RCB	350

In [9]: `# TASK 2 - Zomato`

```
zomato = pd.DataFrame({"city": ["Mumbai", "Mumbai", "Delhi", "Vadodara", "Mumbai"],
                        "order_amount": [500, 700, 400, 600, 300]})

print(zomato.groupby(["city", "restaurant"]).agg({"order_amount": "mean"}))
```

		order_amount
city	restaurant	
Delhi	KFC	400.0
Mumbai	Dominos	600.0
	KFC	300.0
Vadodara	KFC	600.0

In [10]: `# TASK 3 - Flipkart`

```
flipkart = pd.DataFrame({"category": ["Mobile", "Mobile", "Shoes", "Shoes"], "region": ["West", "West", "North", "North"],
                          "sales": [50000, 70000, 30000, 40000], "units_sold": [10, 14, 6, 8]})

print(flipkart.groupby(["category", "region"]).agg({"sales": "sum", "units_sold": "sum"}))
```

		sales	units_sold
category	region		
Mobile	West	120000	12.0
Shoes	North	70000	7.0

In [11]: `# TASK 4 - Revenue`

```
revenue = pd.DataFrame({"region": ["East", "West", "North", "North", "South"], "revenue": [50000, 100000, 60000, 80000, 80000]})

print(revenue.groupby("region").agg({"revenue": ["sum", "max", "min"]}))
```

revenue			
	sum	max	min
region			
East	50000	50000	50000
North	100000	60000	40000
South	80000	80000	80000
West	70000	70000	70000

In [12]: `# TASK 5 - Spotify`

```
spotify = pd.DataFrame({"artist": ["Kahan Kaleshi", "Dhir Khopri", "Kunjan", "Aditya", "Aditya"],
                        "genre": ["Hip-Hop", "Hip-Hop", "Rock", "Rock", "Pop", "Pop"], "streams": [100000, 100000, 100000, 100000, 100000]})

result = spotify.groupby("genre").agg({"streams": ["sum", "mean"]})
```

```
result.columns = ["total_streams", "average_streams"]  
  
print(result.sort_values("total_streams", ascending=False))
```

	total_streams	average_streams
genre		
Pop	4500000	2250000.0
Hip-Hop	2500000	1250000.0
Rock	2000000	1000000.0

In []: